

Vincent Sham



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Skills to Highlight

- **Data Engineering:** 4+ years in architecting enterprise **Data Lakes**, **ETL workflows**, **data modeling**, and **Hadoop** migrations.
- **Financial Tech:** Domain expertise in **Actuarial Science**, **stock market analysis**, and **risk modeling** algorithms.
- **AI & GenAI:** 2 years of experience in **Semi-Supervised Learning**, **LLM orchestration (LangGraph)**, **prompt engineering**, and **RAG pipelines**.
- **Full Stack:** End-to-end development of data applications using **Python (FastAPI/SQL)** and **Next.js/React**.

Software Skills —

Programming:

Python, JavaScript, TypeScript, Java, R, SQL

AI/ML:

scikit-learn, PyTorch, Airflow, XGBoost, LangChain, LangGraph, RAG

Data Engineering:

NumPy, pandas, IBM DataStage, PySpark, Snowflake, dbt, Hadoop

BI Tools:

Power BI, Tableau

Cloud & Systems:

AWS, Linux

DevOps Tools:

Git, Docker, Kubernetes, Terraform

Web Dev:

Next.js, React, Tailwind CSS

Education

2023-2025

MSc in Computer Science

York University

- **Thesis:** Semi-supervised learning with deep generative models for computer vision
- **GPA:** 3.97/4.00

2012-2016

Honors Bachelor of Science

University of Toronto, St. George

- Specialist in Actuarial Science and Major in Statistics
- **GPA:** 3.74/4.00 (High Distinction)

Featured Product

2025

Winsanity: AI-Powered Stock Analysis Agent

Independent Developer

- **Full Stack AI Application | Stack:** Next.js, LangGraph, Vector DB
- **Live Demo:** <https://winsanity.vercel.app/>
- Developed a RAG-driven AI workflow to automate stock analysis, utilizing vector embeddings to query high-density financial data and earnings transcripts.
- Architected end-to-end data pipelines and multi-agent orchestration via LangGraph to analyze unstructured financial data into investment insights.
- Engineered a proprietary "Win Score" algorithm to rank stocks based on fundamental and technical percentile metrics.
- Deployed a high-performance Next.js web application optimized for low-latency financial data visualization.

Experience

2023/04-
2023/09

Machine Learning Engineer Intern

Vector Institute

- Delivered technical demonstrations to industry partners, translating AI and data concepts for business stakeholders.
- Optimized model architecture for business use-cases, improving prediction accuracy on sparse datasets.
- Implemented SOTA Self-Supervised Learning models (TabNet, SAINT) on mixed tabular/time-series data.

2020/01-
2022/12

Product Manager

Silver Team International Textiles Limited

- Translated client business requirements into technical product specifications, managing stakeholder expectations from design to delivery.
- Oversaw the end-to-end product lifecycle, coordinating cross-functional teams to ensure adherence to industry standards.
- Implemented Quality Assurance (QA) protocols and compliance checks to validate product design against specifications.

2018/06-
2019/09

Team Lead - Data Engineering

First North Consulting

- **Client:** Scotiabank - Global Human Resources
- Optimized ETL/ELT performance using IBM DataStage, reducing batch processing times by 80% for high-volume datasets.
- Led an Agile team in designing enterprise-scale ETL data pipelines, migrating on-premise legacy data to Hadoop.
- Architected Master Data Management (MDM) solutions to unify HR demographic data across global branches.
- Coordinated sprint planning and acted as the technical liaison between Product Managers and Developers.

Certificates —

Society of Actuaries Exam P/1
Society of Actuaries Exam FM/2
Society of Actuaries Exam MFE/3F
Society of Actuaries Exam MLC
Society of Actuaries Exam C/4
CFA Level I

2017/04- 2018/06	Data Engineer • Client: Scotiabank - Global Human Resources • Developed robust ETL/ELT workflows to ingest, clean, and transform massive payroll/pension datasets. • Engineered automated data quality checks, data profiling, and cleaning pipelines to ensure integrity of the Landing Zone. • Collaborated with Data Scientists to build feature-rich datasets for downstream analytics. • Partnered with BI teams to optimize Tableau dashboards, improving query performance and load times.	First North Consulting
2016/06- 2016/12	Data Analyst Intern • Engineered SQL scripts for automated Quality Assurance on raw marketing datasets. • Developed VBA macros to model data projections and identify anomalies in historical trends.	Decision Resources Group
2014/09- 2015/04	Teaching Assistant • Conducted tutorials for <i>Intro to Mathematical Proofs</i> , facilitating student understanding of complex logic.	University of Toronto, Mississauga

Academic Projects

2025	Fairness and Bias in AI for Mortgage Approval Graduate Project Stack: Python, LightGBM, SHAP • https://github.com/vincentsham/fairness_aware_lgbm • Built a fairness-aware ML pipeline processing 13M+ HMDA records to predict approvals with minimal bias. • Implemented advanced feature selection (subsampling, leakage detection) to handle high-dimensionality. • Optimized hyperparameters using TPE/Hyperband and dynamic re-sampling, maximizing racial equity.	York University
2025	Stock Sentiment Analysis: News vs. Reddit Graduate Project Stack: NLP, Transformers, Selenium • https://github.com/vincentsham/news_sentiment_analysis • Built automated scrapers (yfinance, PRAW, Selenium) to correlate social sentiment with market volatility. • Deployed LLMs to classify sentiment and categorize fiscal news, identifying leading indicators for stock price. • Conducted time-series analysis (cross-correlation, lead-lag) to quantify information delay in markets.	York University
2024-2025	Semi-Supervised Learning with Hybrid Generative Models Master's Thesis Stack: PyTorch, Generative AI • https://github.com/vincentsham/masters-thesis • Proposed a novel SSL method combining Normalizing Flows with Adversarial Learning to solve mode-collapse. • Developed a hybrid unsupervised loss function, improving decision boundary fitting in low-density regions. • Benchmarked against SOTA generative models, achieving superior conditional sampling performance.	York University